Please check the examination details below before entering your candidate information

Candidate surname	Other names

Centre Number | | | | | |

Candidate Number | | | | |

Pearson Edexcel Level 3 GCE

Paper reference **8MA0/21**

Mathematics

Advanced Subsidiary
PAPER 21: Statistics
Mock Set 2

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided – *there may be more space than you need*.
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Values from statistical tables should be quoted in full. If a calculator is used instead of tables the value should be given to an equivalent degree of accuracy.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 30. There are 4 questions.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

S 7 3 9 7 4 A 0 1 1 2

Pearson

1. A sample of independent values is taken from a discrete random variable X

The sample data are summarised in the grouped frequency table below

x	2 or 3	4	5	6	7	8, 9 or 10
Frequency	20	40	$95 - n$	n	25	15

Outliers for x are defined as any values

$$\text{below } Q_1 - 1.5 \times \text{IQR} \qquad \text{or} \qquad \text{above } Q_3 + 1.5 \times \text{IQR}$$

Given that in these data, $Q_1 = 4$ and $Q_3 = 6$

(a) determine whether there may be outliers in these data.
Give a reason for your answer.

(2)

Given that the median of these data is 5

(b) find the maximum value of n

(3)

A value of x is chosen at random from the sample.

(c) Find the probability that this value is greater than 7

(1)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

Question 1 continued

(Total for Question 1 is 6 marks)

S 7 3 9 7 4 A 0 3 1 2

2. Bea believes that 20% of letters are sent by 1st class post.
Marco claims that less than 20% of letters are sent by 1st class post.

Marco and Bea decide to carry out an appropriate hypothesis test using a 5% significance level.

(a) Write down appropriate null and alternative hypotheses for their test.

(1)

Marco and Bea take a random selection of 25 letters and find that 1 letter was sent by 1st class post.

(b) (i) Showing your working, find the critical region for their test.

(ii) State the conclusion of their test.

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 2 continued

(Total for Question 2 is 5 marks)

3. Edward is practising statistical calculations using values from the large data set.

He selects a random day from June 1987 and calculates the following summary statistics for the Daily Mean Temperature, x °C, for that day from the 8 locations in the large data set.

$$S_{xx} = 133 \qquad \sum x^2 = 2311$$

(a) Using these statistics,

(i) show that the mean of x is 16.5

(3)

(ii) find the standard deviation of x

(2)

Edward now studies the Daily Mean Visibility, v, for the same day in June 1987

He codes the data for 5 locations in the large data set using

$$y = \frac{v}{100} - 20$$

and finds that the mean of y is 10.6 and the standard deviation of y is 12.6

(b) (i) Show that the mean of v is 3060

(2)

(ii) Find the standard deviation of v

(1)

(c) Using your knowledge of the large data set,

(i) state the units for v

(1)

(ii) state, giving a reason, a location in the large data set that Edward could **not** use for his Daily Mean Visibility calculations.

(1)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 3 continued

Question 3 continued

Question 3 continued

(Total for Question 3 is 10 marks)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

S 7 3 9 7 4 A 0 9 1 2

4. A sports club offers its members squash, tennis and a multi-gym.

The club secretary carries out a census to discover how many members play squash, play tennis or use the multi-gym.

(a) Explain what you understand by a census in this context.

(1)

For a randomly selected member of the club, the events S, T and G are defined as

S, the club member plays squash

T, the club member plays tennis

G, the club member uses the multi-gym

The club secretary finds that S and G are independent events.

Given that

$$\mathrm{P}(S) = \frac{3}{4} \qquad \mathrm{P}(T) = \frac{3}{10} \qquad \mathrm{P}(G) = \frac{1}{5}$$

(b) show that for a randomly selected member of the club

$$\mathrm{P}(S \text{ and } G) = 0.15$$

(2)

The club secretary also finds that

- all club members take part in at least one of these three activities
- T and G are mutually exclusive events

(c) (i) Showing your working clearly, find the probability that a randomly selected member of the club plays squash but does **not** play tennis and does **not** use the multi-gym.

(2)

(ii) Draw a Venn diagram to display the three events S, T and G and all of their associated probabilities.

(4)

Question 4 continued

Question 4 continued

(Total for Question 4 is 9 marks)

TOTAL FOR PAPER IS 30 MARKS